Mittuniversitetet
MID SWEDEN UNIVERSITY

Department of
Computer and Electrical Engineering (DET)
Mårten Sjöström (Marten.Sjostrom@miun.se), 010 142 8836
Roger Olsson (Roger.Olsson@miun.se), 010 142 8698

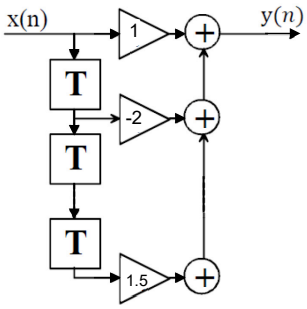
Example examination in

DT081A – Signal and Image Processing, 6 hp

Allowed aids	Dictionary, non-programmable calculator and drawing tools (e.g ruler), Physics Handbook, TeFyMa, and formula collection booklet for mathematics/physics courses at MIUN. A summary of formulas are presented at the end of this exam.
Miscellaneous	Don't forget to: 1. Submit one, or several pages, per solved assignment and NOT several solved assignments per page. 2. Order the handed in pages with the solved assignments in ascending order. 3. Describe and clearly motivate all solutions, assumptions, programming constructs, etc.. 4. Attach all neatly drawn figures that you might want to refer to in a solution. **Failing to follow the above instructions will render point reduction.**

Good Luck!

1. The following block diagram represents a system.



a) Does the system fulfill the LTI properties? Motivate why or why not. (2p)
b) Present the system's difference equation. (3p)

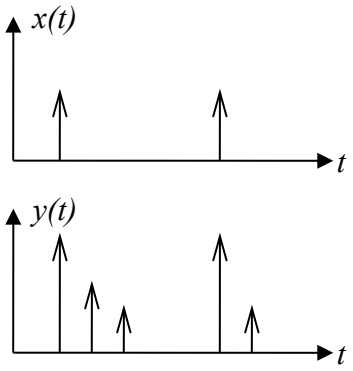
2. Consider a signal $x(t) = 2\cos\left(400\pi t + \frac{\pi}{4}\right) + 4\sin(600\pi t)$.

a) At what frequency f_s must the signal be sampled to be reconstructed with the same frequency content, and what should the cutoff frequency f_c of an ideal reconstruction filter be? (2p)

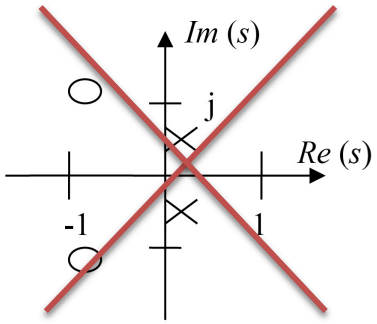
b) Show a graph of the sampled signals spectrum from 0 to $4f_s$. (2p)

3. Which of the following six system representations (*i – vi*) describes a causal and ~~stable~~ LTI-system? For the others not fulfilling the LTI-criterias, motivate what criteria (causal, stabile, linjear, time invariant) that is not filfilled. (6p)

i)



ii)



iii) $y(n-1) - 0{,}1y(n-2) = x(n) - 0{,}2y(n-3)$

iv) $y(n-1) = x(n) - 0{,}6y(n+1)$

v) $h(n) = e^n \sin(0{,}33n)$

vi) $\frac{dy}{dy} + 3\frac{dy^2}{dt} = 2\frac{dx}{dt}$

4. The following image $x(i,j)$ is to be convolved with the kernel $h(i,j)$. Calculate the resulting image $y(i,j)$.

$x(i,j)$

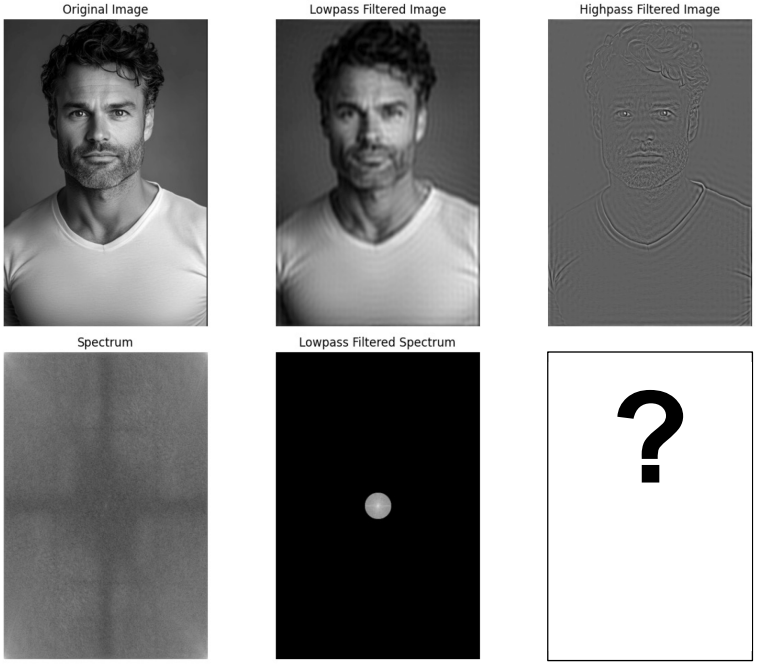
i \ j	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	25	0	0	255	0
2	0	0	0	0	0	0
3	0	0	0	0	50	0
4	0	240	0	0	0	10
5	0	0	0	0	0	0

$h(i,j)$

i \ j	-1	0	1
-1	0.01	0.1	0.01
0	0.1	0.5	0.1
1	0.01	0.1	0.01

5. The below image show how an image is filtered in the spatial frequency domain.

a) Explain how the lowpass image is created
b) How should the spectrum in the region indicated with a question mark be created from the spectrum to produce the highpass filtered image
c) What could be the cause of the ringing effects in the lowpass and filtered images?



6. Compute the 2D Fourier transform for the following 2x2 image block:

$$I = \begin{bmatrix} 64 & 255 \\ 0 & 128 \end{bmatrix}$$